

SECTION 1. IDENTIFICATION OF THE CHEMICAL AND COMPANY

Product name : AQUANOMIC 2.0 LOW TEMP LAUNDRY DETERGENT

Other means of identification : Not applicable

Recommended use : Laundry detergent

Product dilution information : 0,02 % - 0,08 %

Company : Chile ECOLAB S.A.
Avenida Pedro de Valdivia 3801
Santiago, Santiago Chile
(2)-22413300, Telephone: (2)-22381603, Telephone: SAC: 600 241 6600
sac.chile@ecolab.com

Emergency telephone : (+56-2) 2635-3600 (CITUC) EMERGENCY PHONE

Toxicological Emergency Phone : CITUC (+56-2) 2635-3800 (24 hours) Toxicological Emergencies

SECTION 2. HAZARDS IDENTIFICATION

Classification according to NCH 382 : Not classified according to NCh 382

Distintivo según NCh 2190 : Not classified according to NCh 382

GHS Classification
Product AS SOLD

Acute toxicity (Oral) : Category 4
Skin irritation : Category 2
Serious eye damage : Category 1
Skin sensitization : Category 1
Acute aquatic toxicity : Category 2

Product AT USE DILUTION

Not a hazardous substance or mixture.

GHS label elements
Product AS SOLD

Hazard pictograms :



Signal Word : Danger

Hazard Statements : Harmful if swallowed.
Causes skin irritation.
May cause an allergic skin reaction.
Causes serious eye damage.

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Toxic to aquatic life.

Precautionary Statements : **Prevention:**
Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray. Wash skin thoroughly after handling. Do not eat, drink or smoke when using this product. Contaminated work clothing must not be allowed out of the workplace. Avoid release to the environment. Wear protective gloves/ eye protection/ face protection.
Response:
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor. If skin irritation or rash occurs: Get medical advice/ attention. Take off contaminated clothing and wash it before reuse.
Disposal:
Dispose of contents/ container to an approved waste disposal plant.

Product AT USE DILUTION

Precautionary Statements : **Prevention:**
Wash hands thoroughly after handling.
Response:
Get medical advice/ attention if you feel unwell.
Storage:
Store in accordance with local regulations.

Product AS SOLD

Other hazards : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

SECTION 3. COMPOSITION/INFORMATION OF INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
2-methylpentane-2,4-diol	107-41-5	10 - 30
alcohols, c12-16, ethoxylated	68551-12-2	10 - 30
benzenesulfonic acid, dodecyl-, compd. with 2-aminoethanol (1:1)	26836-07-7	10 - 30
Fattyalcohol ethoxylates > C15 and =< 5EO	160875-66-1	10 - 30
fatty acids, c12-18 and c18-unsatd.	90990-15-1	5 - 10
amines, coco alkyltrimethyl, n-oxides	61788-90-7	1 - 5
2-Butenedioic acid (2Z)-, homopolymer	26099-09-2	1 - 5
maleic acid	110-16-7	0,1 - 1
sulfuric acid	7664-93-9	0,1 - 1

Product AT USE DILUTION

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

Product AS SOLD

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use

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a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention.

If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.
Protection of first-aiders	: If potential for exposure exists refer to Section 8 for specific personal protective equipment.
Notes to physician	: Treat symptomatically.

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

Product AS SOLD

Expected acute effects

Eyes	: Causes serious eye damage.
Skin	: Causes skin irritation. May cause allergic skin reaction.
Ingestion	: Harmful if swallowed.
Inhalation	: Health injuries are not known or expected under normal use.

Expected chronic effects

Chronic Exposure	: Health injuries are not known or expected under normal use.
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Product AT USE DILUTION

Expected acute effects

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.

Expected chronic effects

Chronic Exposure	: Health injuries are not known or expected under normal use.
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Most important symptoms / effects

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Irritation, Allergic reactions
Ingestion	: No information available.
Inhalation	: No symptoms known or expected.

Product AT USE DILUTION

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Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

SECTION 5. FIREFIGHTING MEASURES

Product AS SOLD

Extinguishing Agents	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing agents	: None known.
Specific hazards during fire fighting	: Not flammable or combustible.
Products generated in combustion and thermal degradation	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus
Precautions for Emergency Personnel and / or Firefighters	: Use personal protective equipment.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6. MEASURES IN CASE OF ACCIDENTAL SPILLAGE

Product AS SOLD

Personal precautions, protective equipment and emergency procedures	: Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures	: Refer to protective measures listed.
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Environmental precautions	: No special environmental precautions required.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Product AS SOLD

Handling

Advice on safe handling	: Do not ingest. Avoid contact with skin and eyes. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas.
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Storage

Conditions for safe storage	: Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
Recommended and unsuitable packaging by supplier	: Use high density plastic containers.
Storage temperature	: 0 °C to 50 °C

Product AT USE DILUTION

Handling

Advice on safe handling	: Wash hands after handling. For personal protection see section 8.
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Storage

Conditions for safe storage	: Keep out of reach of children. Store in suitable labeled containers.
Recommended and unsuitable packaging by supplier	: Use high density plastic containers.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Ingredients	CAS-No.	Form of exposure	Permissible concentration	Basis
2-methylpentane-2,4-diol	107-41-5	CMP-C	25 ppm	AR OEL
2-methylpentane-2,4-diol	107-41-5	CEIL	25 ppm 121 mg/m3	PE OEL
2-methylpentane-2,4-diol	107-41-5	LB	25 ppm	VE OEL
2-methylpentane-2,4-diol	107-41-5	VLE-P	25 ppm	NOM-010-STPS-2014
2-methylpentane-2,4-diol	107-41-5	Ceiling	25 ppm 125 mg/m3	NIOSH REL
		TWA (Vapor)	25 ppm	ACGIH
		STEL (Vapor)	50 ppm	ACGIH
		STEL	10 mg/m3	ACGIH

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		(Inhalable fraction, Aerosol only)		
sulfuric acid	7664-93-9	CMP	1 mg/m3	AR OEL
		CMP - CPT	3 mg/m3	AR OEL
sulphuric acid	7664-93-9	LPP	0,88 mg/m3	CL OEL
		LPT	3 mg/m3	CL OEL
sulfuric acid	7664-93-9	TWA	1 mg/m3	PE OEL
		STEL	3 mg/m3	PE OEL
sulfuric acid	7664-93-9	LT	0,2 mg/m3	VE OEL
sulfuric acid	7664-93-9	VLE-PPT (Thoracic fraction)	0,2 mg/m3	NOM-010-STPS-2014
sulfuric acid	7664-93-9	TWA (Thoracic fraction)	0,2 mg/m3	ACGIH
		TWA	1 mg/m3	NIOSH REL
		TWA	1 mg/m3	OSHA Z1

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Impervious gloves, resistant to chemicals.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use.
Wash face, hands and any exposed skin thoroughly after handling.
Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Product AS SOLD

Product AT USE DILUTION

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AQUANOMIC 2.0 LOW TEMP LAUNDRY DETERGENT

Appearance	: liquid	liquid
Color	: cloudy, light yellow	colorless
Odor	: mild	not significant
pH	: 4,0 - 5,5, (100 %)	4,0 - 7,5
Flash point	: Not applicable, Does not sustain combustion.	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: No data available	
Evaporation rate	: No data available	
Flammability (solid, gas)	: No data available	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: No data available	
Water solubility	: No data available	
Solubility in other solvents	: No data available	
Partition coefficient: n-octanol/water	: No data available	
Autoignition temperature	: No data available	
Thermal decomposition	: No data available	
Viscosity, kinematic	: No data available	
Explosive properties	: No data available	
Oxidizing properties	: No data available	
Molecular weight	: No data available	
VOC	: No data available	

SECTION 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: Do not mix with bleach or other chlorinated products – will cause chlorine gas.
Conditions to avoid	: None known.
Incompatible materials	: None known.
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus

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AQUANOMIC 2.0 LOW TEMP LAUNDRY DETERGENT

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Toxicity

Product AS SOLD Product

Acute oral toxicity	: Acute toxicity estimate : 1.474 mg/kg
Acute inhalation toxicity	: No data available
Acute dermal toxicity	: Acute toxicity estimate : > 5.000 mg/kg
Skin corrosion/irritation	: No data available
Serious eye damage/eye irritation	: No data available
Respiratory or skin sensitization	: No data available
Carcinogenicity	: Not classifiable as a human carcinogen.
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

Ingredients

Acute inhalation toxicity	: 2-methylpentane-2,4-diol 4 h LC50 Rat: > 1,1 mg/l Test atmosphere: dust/mist
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SECTION 12. ECOLOGICAL INFORMATION

Product AS SOLD Ecotoxicity

Environmental Effects	: Toxic to aquatic life.
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Product

Toxicity to fish	: No data available
Toxicity to daphnia and other aquatic invertebrates	: No data available
Toxicity to algae	: No data available

Ingredients

Toxicity to fish	: alcohols, c12-16, ethoxylated LC50: 1,5 mg/l fatty acids, c12-18 and c18-unsatd. 96 h LC50 Fish: > 3,2 mg/l amines, coco alkyldimethyl, n-oxides 96 h LC50 Fish: 1 mg/l
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AQUANOMIC 2.0 LOW TEMP LAUNDRY DETERGENT

2-Butenedioic acid (2Z)-, homopolymer
96 h LC50 Fish: 580 mg/l

sulfuric acid
96 h LC50: 22 mg/l

Ingredients

Toxicity to daphnia and other aquatic invertebrates : 2-methylpentane-2,4-diol
48 h EC50 Daphnia: 2.800 mg/l

Fattyalcohol ethoxylates > C15 and =< 5EO
48 h Daphnia magna (Water flea): > 1 mg/l

Persistence and degradability

Product AS SOLD

No data available

Product AT USE DILUTION

Readily biodegradable.

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. INFORMATION OF DISPOSAL

Product AS SOLD

- Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.
- Disposal of contaminated packaging : In accordance with what is described by Supreme Decree No. 148, the packaging of the product is considered hazardous waste and must be disposed of through companies authorized to receive and / or treat such waste, which must be issued and final disposal certificate waste.

Product AT USE DILUTION

- Disposal methods : Diluted product can be flushed to sanitary sewer.
- Disposal considerations : Dispose of in accordance with local, state, and federal regulations.
- Disposal of contaminated packaging : In accordance with what is described by Supreme Decree No. 148, the packaging of the product is considered hazardous waste and must be disposed of through companies authorized to receive and / or treat

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY DETERGENT

such waste, which must be issued and final disposal certificate waste.

SECTION 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

Official transport classification of the United Nations

Not dangerous goods

Air transport (IATA)

Official transport classification of the United Nations

Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

Official transport classification of the United Nations

Not dangerous goods

SECTION 15. REGULATORY INFORMATION

Registrations and Certifications

Chile: Our HDS meets Chilean Normative: NCH 2245: 2015 (Safety Data Sheet for chemical products - Content and order of sections)

The receiver should pay attention to the possible existence of local regulations.

NCh 1411: Prevention of risks, IV identification of risks of materials

NCh 2190: Transportation of Hazardous Substances - Pictograms for hazard identification

NCh 382: Dangerous Good - Classification

D.S. No. 594: Minimum basic conditions in the workplace

D.S. No. 148: Hazardous waste disposal

D.S. No. 72: Mining Safety Regulations

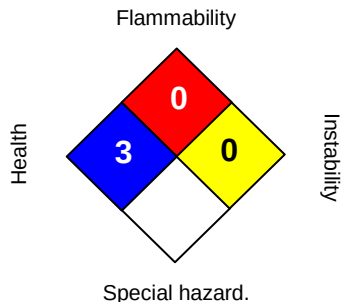
D. S. No. 43: reports on the storage of hazardous substances

D. S. No. 40: Reporting on exposure hazards

SECTION 16. OTHER INFORMATION

Product AS SOLD

NFPA:



HMIS III:

HEALTH	3*
FLAMMABILITY	0
PHYSICAL HAZARD	0

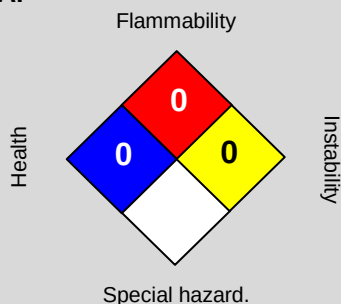
0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

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AQUANOMIC 2.0 LOW TEMP LAUNDRY DETERGENT

NFPA:



HMIS III:

HEALTH	0
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 04.10.2018
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Prepared by : Regulatory Affairs

While there is no change in the formula or hazard classifications, this SDS remains valid.

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.